

Attention-Enhanced LSTM Autoencoder for Login Anomaly Detection

Hizkia

Graduate Informatics Program
Universitas Pelita Harapan
Jakarta, Indonesia
01679240005@student.uph.edu

Pujianto Yugopuspito

Graduate Informatics Program
Universitas Pelita Harapan
Jakarta, Indonesia
yugopuspito@uph.edu

Abstract—Seeing the development of the times, organizations and companies are very dependent on IT systems that can help daily business processes. The IT systems used are often exposed to various cybersecurity threats. One of the threats experienced is unauthorized access through login attacks, which can cause data leakage to disrupt the IT system used and cause the company or organization to suffer losses. Login logs are critical in one of the steps to monitor and find anomalies in user login activities. Still, the large volume and complexity of login logs make it difficult to find anomalies from user login activities if done manually.

To address these challenges, this study proposes an attention-enhanced long short-term memory (LSTM) autoencoder for detecting anomalous login activities. The attention mechanism improves temporal pattern learning and highlights critical timesteps, providing both detection accuracy and interpretability. Using a dataset of 6,314 login records from 60 users, the proposed model achieved stable convergence and competitive anomaly detection performance, with an F1-score of 0.81 for anomalous sequences, slightly outperforming the baseline LSTM autoencoder (0.80). Importantly, attention weight visualizations revealed that anomalous sequences consistently received higher focus in later timesteps, validating the interpretability of the approach. These findings demonstrate that integrating attention into autoencoders not only enhances detection capability but also contributes to transparency in anomaly detection, an essential factor in security-critical environments.

Index Terms—Long Sort Term Memory, Attention Mechanism, Anomaly Detection, Identity and Access Management

I. INTRODUCTION

Cybersecurity is crucial in corporate environments due to the reliance on information systems. Cyber threats like malware, phishing, and ransomware increasingly target login credentials via credential stuffing, brute force attacks, and session hijacking [1]. Login attacks and unauthorized access remain major threats to enterprise networks. Traditional intrusion detection systems often miss credential-based lateral movement, advanced methods are needed [2]. This is a part of identity and access management. Visualization tools such as APT-Hunter

can help analysts identify malicious log-in patterns [3], and the modeling of the network login structure with anomaly detection has shown effectiveness, achieving a 82% detection rate with a 0.3% false positive rate in a real-world environment [2]. Continuous authentication and behavioral biometrics offer additional protection against login attacks in web applications [4].

Login logs are essential for detecting login anomalies, but their volume and complexity hinder manual analysis. First, label the log record as normal or not. Labeling is a potentially time-consuming, error-prone, and tedious process when performed with supervised techniques [5]. Supervised techniques cannot also detect unknown attacks because these techniques rely on prior knowledge [6]. Large-scale monitoring of log-in activity faces challenges such as high data volumes, time constraints for manual analysis, and a shortage of cybersecurity experts. Many systems still rely on rule-based detection, which tends to be ineffective in detecting new attacks (zero-day threats) or complex anomalies [7].

Although most logs are instructed, log data also includes artifacts that point to malfunctions or events. As a result, anomaly detection methods that seek to automatically reveal unusual or otherwise pertinent system behavior patterns are frequently evaluated using data logs [8]. This study employs anomaly detection techniques to identify login attacks and unauthorized access patterns. Login attacks involve unauthorized access to systems using legitimate accounts. Anomaly detection establishes a baseline of normal system behavior and flags significant deviations as anomalies, which may indicate attacks or malfunctions. The system learns normal behavior during a training phase and continuously updates its baseline profiles.

The LSTM component excels at handling long-term dependencies within sequential data, allowing the model to capture extended behavioral patterns and trends that characterize normal system operation [9]. The autoencoder architecture complements this capability by learning to reconstruct input data from the normal class, with

reconstruction error serving as the primary indicator of anomalous behavior. These neural networks are trained to minimize the difference between original input and reconstructed output, incorporating this loss of reconstruction into the training objective [10].

The LSTM Autoencoder combines two deep learning models: Long Short-Term Memory (LSTM) and autoencoders. LSTM is effective in capturing temporal dependencies in sequential data [11], capable of learning over extended time gaps by using gated mechanisms that control the flow of error signals through time [12]. The autoencoder, consisting of an encoder, code, and decoder, reconstructs input data and calculates reconstruction error using a loss function to compare outputs with the original input [13]. When trained on normal data, if the reconstruction error exceeds a learned threshold, the input is flagged as an anomaly [14], [15]. Additionally, autoencoders can capture non-linear correlations and serve as a tool for dimensionality reduction [16].

The attention mechanism enhances the model's ability to selectively focus on critical portions of the input sequence. By generating a context vector that emphasizes the most salient features of the latent representation, the attention layer enables the decoder to concentrate on the most informative elements of the attended vector. This selective focus results in significantly reduced reconstruction loss compared to standard approaches, as the decoder can prioritize the most relevant temporal information as the attended vector [17].

Unsupervised learning is a machine learning approach used to discover patterns and relationships in unlabeled data. Unlike traditional distance-based and density-based anomaly detection methods, unsupervised techniques can accurately detect small anomalies in cyclic patterns [18].

This research investigates anomaly detection in login activity logs by comparing a standard LSTM autoencoder with an attention-enhanced variant. While prior research has widely adopted LSTM autoencoders for anomaly detection across domains such as network intrusion detection [19], system log analysis [20], and IoT security [21]. These models leverage temporal dependencies to learn normal behavioral patterns and detect deviations through reconstruction errors, showing promising detection accuracy in both labeled and unlabeled scenarios. Most of the research have focused primarily on reconstruction accuracy without addressing the interpretability of detected anomalies. Such limitations reduce their applicability in corporate security contexts, where understanding why an anomaly occurs is as critical as detecting it. Unlike prior work, our model improves interpretability by using attention to highlight critical time steps, addressing a key limitation of previous LSTM autoencoder approaches.

II. METHODOLOGY

The proposed approach processes time-series login data obtained from the identity provider, transforming raw sign-in logs into structured sequences for model training. in several stages: data collection, data preprocessing, feature selection, and cleaning. The autoencoder learns normal behavior patterns and uses reconstruction error to identify deviations indicative of potential security anomalies.

A. Data Collection

Login logs from the Microsoft Entra ID interactive sign-in, covering the last two months, will be exported for data collection. Logins from interactive identity providers, covering the last two months. The first month is for training and the second month is for test data. This logs are representing authentication activities of 60 users accessing the company's internal applications. These applications are part of the daily operational workflow. Each record includes attributes such as IP address, location, browser, operating system, login status, failure reason, multifactor authentication (MFA) result, latency, timestamp, temporal features (hour, day of week, weekend flag, and inter-event time delta) were derived to capture login behavior patterns.

B. Data Preprocessing

Data preprocessing was carried out in multiple stages to prepare the login logs for anomaly detection. From the Date Time column, several derived features were generated to capture temporal patterns: Hour (0–23) indicating login time, DayOfWeek (0 = Monday, 6 = Sunday) to track weekday activity, IsWeekend to distinguish weekday versus weekend logins, TimeDelta representing the interval between consecutive logins, and Timestamp as the Unix time in seconds. These features help identify time-specific and behavioral login patterns for analysis.

The next step to be taken is selecting features to be used; the purpose of this step is to select the relevant columns for anomaly detection. Here is a list of the columns that will be used: *IP address, Location, Operating system, Browser, Status, Failure reason, Multifactor authentication result, Latency, Hour, DayOfWeek, IsWeekend, TimeDelta, and Timestamp.*

After selecting features, the subsequent step involves handling missing values. Since only categorical data is affected, missing values are replaced with "unknown." For instance, a missing browser type will be labeled "unknown browser."

To separate normal and anomalous behavior, filtering rules were applied: (1) login hour outliers (via IQR), (2) non-Indonesia locations, (3) failure reasons other than "Other", (4) MFA results outside the normal set (MFA

satisfied by token, MFA completed in Azure AD, MFA satisfied by strong authentication), (5) Unknown OS or Unknown Browser, and (6) users employing more than K IP addresses in a 24-hour window. A login was labeled anomalous if at least two conditions were satisfied, except for location outside Indonesia, which was always treated as anomalous. This labeling strategy ensures that anomalies reflect multi-faceted deviations from expected login behavior, while simultaneously addressing high-risk cases such as foreign access attempts.

Categorical features (IP, location, browser, operating system, status, failure reason, MFA result) were converted into integer IDs using hash encoding, while numerical features (latency, hour, day of week, weekend flag, time delta) were normalized with Min–Max scaling. Sequences of length 10 were constructed to represent temporal login patterns.

C. Model Architecture

Fig. 1 illustrates the architecture of the LSTM autoencoder model with an attention mechanism for anomaly detection. The model consists of three main components: the LSTM Encoder, the Attention Mechanism, and the LSTM Decoder. The original LSTM autoencoder lacked an attention mechanism. The attention mechanism improves the autoencoder by focusing on key hidden states, enhancing interpretability via sequence visualization [22]. This mechanism also helps maintain accuracy as input sequence lengths increase, reducing model error in time series tasks [23].

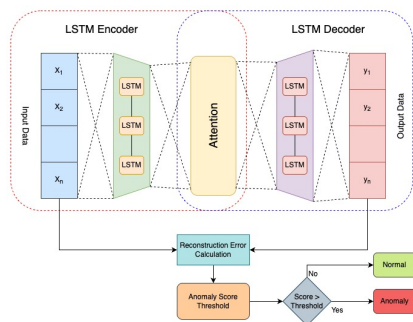


Fig. 1. LSTM Autoencoder with Attention Model Architecture

The proposed model is an attention-enhanced LSTM autoencoder designed for anomaly detection in sequential login log data. The input to the model is a sequence of fixed length $T = 10$, where each timestep is represented by a feature vector of size $F = 12$.

The *encoder* is composed of two stacked unidirectional LSTM layers. The first layer contains 96 hidden units with ℓ_2 regularization ($\lambda = 0.002$) and outputs the full sequence of hidden states. A dropout rate of 0.2 is applied to reduce overfitting. To enhance focus

on relevant temporal features, an attention mechanism is applied after the first LSTM layer. The attention block computes a softmax distribution over the temporal dimension, generates weights, and applies them via element-wise multiplication with the LSTM outputs. This enables the model to assign higher importance to informative timesteps. The attended sequence is then passed into a second LSTM layer with 48 hidden units, producing a fixed-length latent representation.

The decoder mirrors the encoder, reconstructing the input sequence from the latent vector. The representation is first repeated T times using a RepeatVector layer, followed by two unidirectional LSTM layers with 48 and 96 hidden units, respectively, both with ℓ_2 regularization and a dropout rate of 0.2. Finally, a TimeDistributed dense layer with F units is used to reconstruct the feature vectors at each timestep.

The model is trained using the Adam optimizer (learning rate = 0.0005) with mean squared error (MSE) as the loss function. In order to assess the model's performance, the accuracy metric is also supplied [24]. Early stopping with a patience of 15 epochs is applied to prevent overfitting and retain the best weights.

D. Training Procedure

The training process involves preparing time-series login data and training two models: an LSTM autoencoder and an LSTM autoencoder with an attention mechanism. Both models are trained to reconstruct input sequences and detect anomalies based on reconstruction error, allowing for a comparative evaluation of their performance.

E. Anomaly Detection Mechanism

Anomalies are identified by computing reconstruction errors between input and output sequences. A threshold is selected using validation data by maximizing the F1-score, and sequences exceeding this threshold are flagged as anomalous. For interpretability, attention weights are visualized to indicate which timesteps contributed most to the reconstruction error.

III. RESULTS AND DISCUSSION

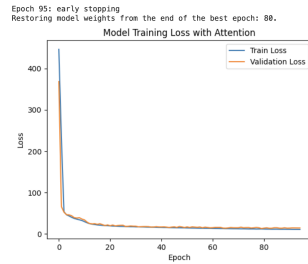
The LSTM autoencoder with the attention mechanism was trained using 3,435 sequences of login activity data and validated on 859 sequences. The model architecture consisted of multiple LSTM layers with dropout and attention applied to the encoder's output. The total number of trainable parameters was 145,261, as seen in Table I.

Figure 2 shows the training results of both models. The attention-based autoencoder converged quickly, reaching stable loss around epoch 20. Early stopping restored the best weights from epoch 80.

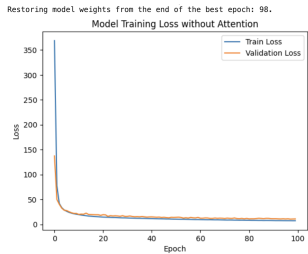
The baseline model without attention needed more time, with best weights at epoch 98. It also showed a

TABLE I
LSTM AUTOENCODER AND ATTENTION LAYER

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 10, 12)	0	—
lstm (LSTM)	(None, 10, 96)	41,856	input_layer[0][0]
dropout (Dropout)	(None, 10, 96)	0	lstm[0][0]
dense (Dense)	(None, 10, 1)	97	dropout[0][0]
flatten (Flatten)	(None, 10)	0	dense[0][0]
activation (Activation)	(None, 10)	0	flatten[0][0]
repeat_vector (RepeatVector)	(None, 96, 10)	0	activation[0][0]
permute (Permute)	(None, 10, 96)	0	repeat_vector[0][...]
multiply (Multiply)	(None, 10, 96)	0	dropout[0][0], permute[0][0]
lstm_1 (LSTM)	(None, 48)	27,840	multiply[0][0]
repeat_vector_1 (RepeatVector)	(None, 10, 48)	0	lstm_1[0][0]
lstm_2 (LSTM)	(None, 10, 48)	18,624	repeat_vector_1[...]
dropout_1 (Dropout)	(None, 10, 48)	0	lstm_2[0][0]
lstm_3 (LSTM)	(None, 10, 96)	55,680	dropout_1[0][0]
time_distributed (TimeDistributed)	(None, 10, 13)	1,164	lstm_3[0][0]
Total params: 145,261 (567.43 KB)			
Trainable params: 145,261 (567.43 KB)			



(a) with Attention



(b) without Attention

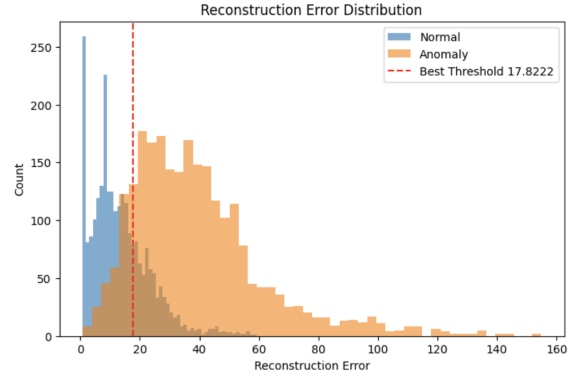
Fig. 2. Training and Validation Loss Curve of LSTM Autoencoder

slightly larger gap between training and validation loss. This means attention helps the model train faster and generalize better by focusing on key temporal features.

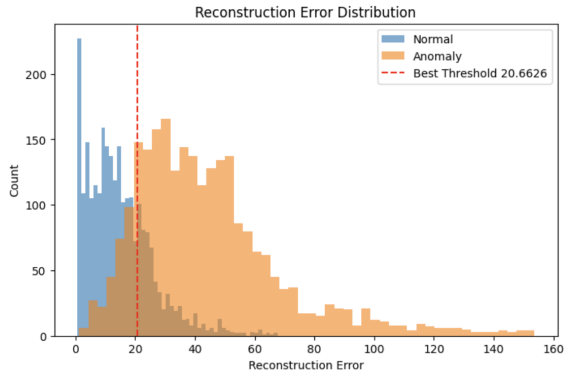
A. Selection of Anomaly Threshold (Best F1 Search)

To determine the decision boundary between normal and anomalous sequences, we applied a threshold search across the reconstruction error distribution. The optimal threshold was selected by maximizing the F1-score against ground-truth labels, ensuring a balance between anomaly detection and false alarms.

Fig. 3 illustrates the reconstruction error distributions of normal (blue) and anomalous (orange) sequences, with the optimal threshold shown as a red dashed line. Normal data clusters tightly around low error values, while anomalous sequences extend toward higher errors, confirming that the LSTM autoencoder learns normal patterns but fails to reconstruct anomalous inputs, resulting in larger errors.



(a) without attention



(b) with attention

Fig. 3. Reconstruction Error Distribution

In the baseline model without attention Fig. 3(a), the best threshold was approximately 17.82, separating most normal and anomalous samples. However, some overlap near the boundary remained. The attention-enhanced model Fig. 3(b) achieved a clearer separation, with the threshold increasing slightly to 20.66. The attention mechanism shifted anomalous errors further into the higher range while keeping normal errors tightly clustered, thereby reducing overlap and improving discrimination.

Detection performance at the selected thresholds was further validated using classification reports and confusion matrices, ensuring both interpretability and quantitative evaluation of the models.

B. Evaluation Metrics Analysis

Table II presents the performance of the LSTM autoencoders, comparing the baseline and the attention-enhanced model. For anomaly detection, both achieved strong recall (0.87), showing effectiveness in identifying anomalies, with the attention model slightly improving precision (0.76 vs. 0.75) and F1-score (0.81 vs. 0.80). For normal detection, precision was higher (0.85) but recall dropped (0.72 baseline, 0.74 with attention), leading

to lower F1-scores (0.78 and 0.79). Overall, both models were better at detecting anomalies than normals, with attention providing small but consistent improvements across all metrics.

TABLE II
EVALUATION METRICS OF LSTM AUTOENCODER MODELS

Model	Precision	Recall	F1-score	Detect
LSTM Autoencoder	0.75	0.87	0.80	Anomaly
LSTM Autoencoder + Attention	0.76	0.87	0.81	Anomaly
LSTM Autoencoder	0.85	0.72	0.78	Normal
LSTM Autoencoder + Attention	0.85	0.74	0.79	Normal

C. Heatmap Visualization and Average Attention Results

To better understand the model's behavior, we visualized the attention-enhanced LSTM autoencoder's reconstruction errors. For anomalous cases Fig. 4(a), errors were higher and concentrated in features such as Failure Reason, Multifactor Authentication, and IP Address, especially between timesteps 6 and 8, indicating difficulties in reconstructing irregular behaviors like failed authentications or sudden IP changes. In contrast, the heatmap for normal logins Fig. 4(b) showed lower, more evenly distributed errors, with only minor variations in Location and Browser. These results demonstrate that the model effectively distinguishes normal from anomalous login activity by capturing both feature-specific irregularities and temporal deviations.

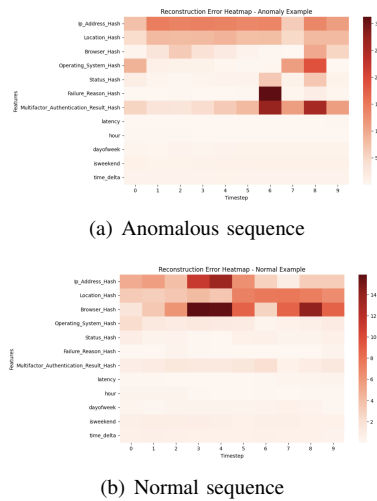


Fig. 4. Heatmap Results

Fig.5 illustrates the average attention distribution across timesteps for normal and anomalous login sequences. Normal sequences show a sharp rise in attention within the first three timesteps, stabilizing around timestep five and maintaining consistently high values, indicating the model's ability to quickly capture and focus on informative temporal dependencies in typical login behavior. In contrast, anomalous sequences display

a slower increase in attention that stabilizes at a lower magnitude, reflecting disrupted temporal associations and weaker attention allocation. This difference suggests that anomalies hinder the model's capacity to establish strong temporal focus, reinforcing their irregular and unpredictable nature.

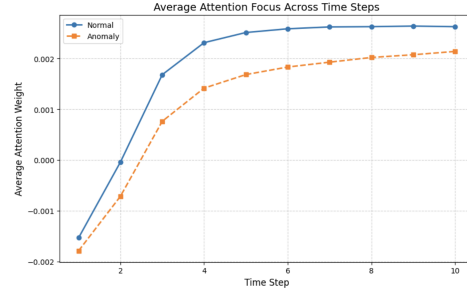


Fig. 5. Average Attention.

These visualizations highlight the model's ability to differentiate between normal and abnormal patterns based on where the attention mechanism is focused during reconstruction. The contrast in attention distributions offers interpretability and supports the model's anomaly detection decisions.

D. Feature Values Results

Without attention, the model lacks introspection. While we can visualize feature values for anomalous sequences (Fig.6), identifying significant time steps remains impossible, hindering explainability and transparency—critical for real-world applications.

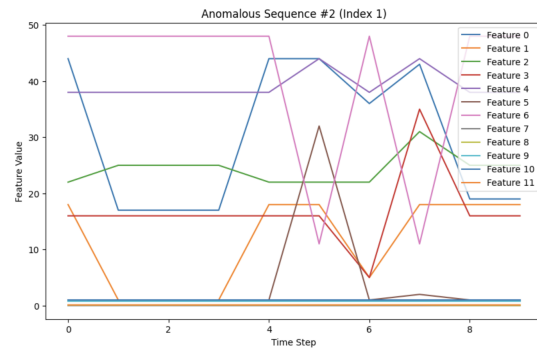


Fig. 6. Feature values.

However, this purely statistical approach lacks insight into the causes of anomalies. Without attention mechanisms, we can only observe overall reconstruction error, hindering our ability to pinpoint the time steps or features most responsible for an anomaly.

IV. CONCLUSION AND FUTURE WORK

This study proposed an attention-enhanced LSTM autoencoder for detecting anomalous login behavior.

The integration of attention not only improved temporal pattern learning but also enhanced interpretability by highlighting influential timesteps. Experiments showed that the model converges reliably and effectively distinguishes normal and anomalous logins based on reconstruction error. Attention visualizations confirmed that anomalies attract greater focus at later timesteps, reflecting irregular login dynamics.

Quantitative results support these findings: the attention-based model achieved an F1-score of 0.81 for anomalies and 0.79 for normal detection, slightly outperforming the baseline model (0.80 and 0.78, respectively). While performance gains are modest, attention introduces important interpretability benefits, allowing the identification of specific temporal points linked to anomalies. This makes the approach particularly valuable in cybersecurity contexts where both accuracy and transparency are critical.

Future Work will focus on several enhancements:

- Online/Real-Time Detection: Extend the model to support streaming data for real-time anomaly detection and alerting.
- Model Generalization: Testing the framework on different datasets and identity management systems to ensure robustness and adaptability.

By advancing both detection capability and interpretability, this research contributes to more secure and transparent access control systems.

REFERENCES

- [1] A. Waheed, M. A. Shah, and A. Khan, "Secure login protocols: An analysis on modern attacks and solutions," in *2016 22nd International Conference on Automation and Computing (ICAC)*, Colchester, UK, 2016, pp. 535–541.
- [2] H. Siadati and N. D. Memon, "Detecting structurally anomalous logins within enterprise networks," *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security*, 2017. [Online]. Available: <https://api.semanticscholar.org/CorpusID:19628082>
- [3] H. Siadati, B. Saket, and N. D. Memon, "Detecting malicious logins in enterprise networks using visualization," *2016 IEEE Symposium on Visualization for Cyber Security (VizSec)*, pp. 1–8, 2016. [Online]. Available: <https://api.semanticscholar.org/CorpusID:5679763>
- [4] R. U. Rahman and D. S. Tomar, "Taxonomy of login attacks in web applications and their security techniques using behavioral biometrics," in *Modern Theories and Practices for Cyber Ethics and Security Compliance*, W. Y. et al., Ed. Hershey, PA: IGI Global Scientific Publishing, 2020, pp. 122–139.
- [5] M. S. Elsayed, N. A. Le-Khac, S. Dev, and A. D. Jurcut, "Network anomaly detection using lstm based autoencoder," in *Q2SWinet 2020 - 16th ACM Symposium on QoS and Security for Wireless and Mobile Networks*, 2020, pp. 37–45.
- [6] Z. Zhao, C. Xu, and B. Li, "A lstm-based anomaly detection model for log analysis," *Journal of Signal Processing Systems*, 2021.
- [7] K. Thakur, S. Kopecky, M. Nuseir, M. L. Ali, and M. Qiu, "An analysis of information security event managers," in *3rd IEEE Int. Conf. on Cyber Security and Cloud Computing (CSCloud 2016) and 2nd IEEE Int. Conf. of Scalable and Smart Cloud (SSC 2016)*, 2016, pp. 210–215.
- [8] M. Landauer, F. Skopik, and M. Wurzenberger, "A critical review of common log data sets used for evaluation of sequence-based anomaly detection techniques," *Proc. ACM Softw. Eng.*, vol. 1, no. FSE, Jul. 2024. [Online]. Available: <https://doi.org/10.1145/3660768>
- [9] F. E. Laghrissi, S. Douzi, K. Douzi, and B. Hssina, "Intrusion detection systems using long short-term memory (lstm)," *Journal of Big Data*, vol. 8, no. 1, 2021.
- [10] A. Goodge, B. Hooi, S. K. Ng, and W. S. Ng, "Robustness of autoencoders for anomaly detection under adversarial impact," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, C. Bessiere, Ed. International Joint Conferences on Artificial Intelligence Organization, 7 2020, pp. 1244–1250, main track. [Online]. Available: <https://doi.org/10.24963/ijcai.2020/173>
- [11] B. Sharma, P. Pokharel, and B. Joshi, "User behavior analytics for anomaly detection using lstm autoencoder-insider threat detection," in *ACM International Conference Proceeding Series*, 2020.
- [12] S. Hochreiter and J. U. Schmidhuber, "Long short-term memory," *Neural Computation*, 1997, original LSTM paper.
- [13] H. C. Altunay, Z. Albayrak, and M. Çakmak, "Autoencoder-based intrusion detection in critical infrastructures," *Current Trends in Computing*, vol. 2, no. 1, p. 1–12, 2024. [Online]. Available: <https://dergipark.org.tr/en/pub/ctc/issue/86213/150005>
- [14] Y. Wei, J. Jang-Jaccard, F. Sabrina, W. Xu, S. Camtepe, and A. Dunmore, "Reconstruction-based lstm-autoencoder for anomaly-based ddos attack detection over multivariate time-series data," 2023. [Online]. Available: <http://arxiv.org/abs/2305.09475>
- [15] S. Maleki, S. Maleki, and N. R. Jennings, "Unsupervised anomaly detection with lstm autoencoders using statistical data-filtering," *Applied Soft Computing*, vol. 108, 2021.
- [16] Z. Chen, C. K. Yeo, B. S. Lee, and C. T. Lau, "Autoencoder-based network anomaly detection," in *2018 Wireless Telecommunications Symposium (WTS)*, 2018, pp. 1–5.
- [17] A. Oluwasanmi, M. U. Aftab, E. Baagyer, Z. Qin, M. Ahmad, and M. Mazzara, "Attention autoencoder for generative latent representational learning in anomaly detection," *Sensors*, vol. 22, no. 1, 2022.
- [18] M. Munir, S. A. Siddiqui, A. Dengel, and S. Ahmed, "Deepant: A deep learning approach for unsupervised anomaly detection in time series," *IEEE Access*, vol. 7, pp. 1991–2005, 2019.
- [19] *Proceedings of the 26th IEEE Signal Processing and Communications Applications Conference (SIU)*. Izmir, Turkey: IEEE, 2018.
- [20] H. Homayouni, S. Ghosh, I. Ray, S. Gondalia, J. Duggan, and M. G. Kahn, "An autocorrelation-based lstm-autoencoder for anomaly detection on time-series data," in *Proceedings of the 2020 IEEE International Conference on Big Data (Big Data 2020)*. IEEE, Dec. 2020, pp. 5068–5077.
- [21] A. Pinto, L. C. Herrera, Y. Donoso, and J. A. Gutierrez, "Enhancing critical infrastructure security: Unsupervised learning approaches for anomaly detection," *International Journal of Computational Intelligence Systems*, vol. 17, no. 1, pp. 1–20, 2024.
- [22] J. Pereira and M. Silveira, "Unsupervised anomaly detection in energy time series data using variational recurrent autoencoders with attention," in *17th IEEE International Conference on Machine Learning and Applications (ICMLA)*, 2018, pp. 1275–1282.
- [23] T. Tayeh, S. Aburakhia, R. Myers, and A. Shami, "An attention-based convlstm autoencoder with dynamic thresholding for unsupervised anomaly detection in multivariate time series," *Machine Learning and Knowledge Extraction*, vol. 4, no. 2, pp. 350–370, 2022.
- [24] F. Lachekhab, M. Benzaoui, S. A. Tadjer, A. Bensmaine, and H. Hamma, "Lstm-autoencoder deep learning model for anomaly detection in electric motor," *Energies*, vol. 17, no. 10, 2024.